Your 3-4 hour revision route

Applied Analytics in Business | COS10082 | Compact guide

Use this guide with your 120-question practice set. Read a section, cover it, explain the ideas aloud, then check yourself. Source references are pages 1-208 in the original revision PDF viewer, rather than the page numbers printed inside separate handouts.

A complete 3-hour session

Minutes	Task	Guide pages
0-10	Orient yourself; identify unfamiliar terms.	1
10-30	Learn definitions, drivers, benefits and big data.	2-3
30-50	Reproduce the four-capability table from memory.	4
50-75	Review development, sourcing, customization and vendors.	5-6
75-110	Review management, data quality, governance and strategy.	7-9
110-145	Work through Power Query, Tableau, joins and cleaning.	10-12
145-170	Attempt 15-20 shuffled MCQs; cover their answers.	Practice set
170-180	Review mistakes and recall the lists below.	This guide

If you have 4 hours

Add 10 minutes of breaks, 35 minutes for another 20-25 MCQs and reviewing errors, and 15 minutes to revisit your weakest section. Total: 240 minutes. These are suggested study timings, not predicted exam weightings.

Final 10-minute recall checklist

- Data -> information -> knowledge; BI product vs process vs tool vs solution.
- Four capabilities, enabling technologies and customization rules.
- Plan -> Analyse -> Design -> Build -> Deploy -> Review; Waterfall vs Agile.
- Six data-quality dimensions; four vendor categories; governance vs BICC.
- Five strategy areas; two meanings of BI centralization.
- UnitPrice x Quantity; group by product AND year; merge then expand.
- Blue vs green; dimension vs measure; chart purpose; union vs join vs blending.

Revision method: explain the difference between similar terms. For scenario MCQs, identify the missing function or the precise failure before choosing an answer.

Study schedule and recall method are suggested for this guide. Source content: original PDF pp. 8-140 and 146-193. Software terminology follows the revision notes.

1. Core definitions and distinctions

Learn the meaning, then recognize it in a scenario.

Term	Meaning	Recognition example
Data	Recorded values or facts; may be incorrect.	161.2, 175.3, 166.4 without context.
Information	Data with context, relevance and purpose.	Those values are students' heights.
Knowledge	Understood or justified relationships that can be applied.	Use sales and bread stock to decide how much bread to order.

Source: PDF pp. 8-9.

Business intelligence (BI) supplies useful information and knowledge for decision makers using internal/external sources and structured/unstructured information. The notes identify knowledge as a key output, and also describe user-friendly information and new insights as BI outputs.

BI term	Remember
Product	The useful information and knowledge produced.
Process	Obtaining, analysing and distributing information/knowledge.
Tool	Application software for access, analysis and presentation.
Solution	The deployed combination of tools and sources supporting the process; an organization may use several solutions.

Source: PDF pp. 11, 13, 23, 41, 71, 85.

BI and related technologies

- Data warehouse: one logical repository for organizational data.
- Data mining: discover hidden patterns in electronically stored data.
- Decision support system (DSS): combine data and prior knowledge to create rules guiding decisions.
- BI: connect these resources to useful information and knowledge, including presentation to people with limited technical expertise.
- Knowledge management (KM): create, convert, capture, share and apply knowledge. Unlike BI, data is not its main concern except in knowledge discovery. BI results can become KM inputs.

Source: PDF pp. 23, 25, 27-28.

Tacit knowledge = undocumented experience or intuition, difficult to store/transmit. Explicit knowledge = documented or encoded, such as a manual, financial statement or lessons-learned record.

Source: PDF p. 24.

MCQ traps: a warehouse stores data; mining finds patterns; BI is broader than either. A software purchase is a tool, not automatically a complete deployed solution. Matching numbers alone do not establish useful knowledge.

Cover and recall: explain why daily sales figures are information, but a bread-ordering decision based on sales and stock demonstrates applied knowledge.

2. Drivers, benefits, big data and cases

Use the business problem to identify the BI contribution.

Driver	Why BI is needed
Exploding data volumes	Cheaper storage, electronic connections and regulatory changes increase stored data.
Complicated decisions	Global competition and diverse structured/unstructured sources add factors to consider.
Quick reflexes	Volatile markets shorten the time available to act.
Technological progress	DSS, ERP, warehousing, data mining and text mining enable BI.

Three delays constrain quick decisions: converting source data, integrating information across sources, and making results available to decision makers. A hotel manager deciding within minutes whether to accept a room bid illustrates quick reflexes.

Source: PDF pp. 16-22.

Contributions and organizational benefits

Four contributions: user-friendly real-time information; new knowledge from the past; responsive and anticipative decisions using current information and predictions; improved future planning.

- Operational performance: identify what needs improvement. Example: on-time gas deliveries but invoices a week late; the company required invoicing within a day.
- Customer service: find problems and solutions, reduce concerns and improve retention. Example: identify mobile customers likely to change carrier.
- New opportunities: discover previously unknown patterns and track innovative projects.

Source: PDF pp. 34-36.

Big data and analytics

3 Vs: volume = amount; variety = different forms; velocity = generation speed. Examples include video, sensor readings, geospatial data, messages and machine logs. Analytics is a subset of BI covering statistics, forecasts, models and simulations. Big-data analytics supplements BI/traditional analytics; big data is another input source, not a replacement for BI.

Source: PDF pp. 64-67.

Case recognition

Case	Problem -> BI result
Tesco	Rewards-card purchase data -> customer categories and tailored offers. Redemption about 20% vs 2% industry norm: roughly 10 times the rate.
St. Joseph Medical Center	Analyse patient reports -> identify potential problems earlier; examine nurse response times; provide information to authorized staff.
PhoneWorks / LucidEra	Static current Salesforce reports lacked historical analysis -> track average deal size, days to close and stage-by-stage bottlenecks.

Source: PDF pp. 15, 142, 144. Case figures and product descriptions are as reported in the notes.

Cover and recall: a missed market opportunity caused by slow report delivery illustrates which driver and which of the three delays? Answer: quick reflexes; making results available.

3. The four BI capabilities

Memory -> Integration -> Insight -> Presentation

Capability	Function / why needed	Enabling technologies
Organizational memory	Preserve accumulated data, information and explicit knowledge. Needed as stored history, electronic content and repositories grow.	Data warehouses; ERP; knowledge repositories; digital-content and document management.
Information integration	Link and synthesize historical/current, internal/external, structured/unstructured content. Needed for disconnected systems and diverse sources.	Environmental scanning; text mining; web mining; RFID.
Insight creation	Find patterns, trends and relationships; describe, understand and predict. Needed for complex decisions and quick responses.	Data mining; business analytics; real-time decision support.
Presentation	Deliver useful information to users in an appropriate form. Needs vary by role, task and individual preference.	OLAP; visualization; dashboards; scorecards; business performance management.

Source: PDF pp. 43, 46-62.

Trace an entire scenario

Stored purchase history = memory. Add live sales, customer comments and competitor reports = integration. Discover demand patterns or forecast demand = insight. Show a tailored management dashboard = presentation. Integration supplies synthesized content for insight creation; presenting data does not automatically discover new patterns.

Source: PDF pp. 43, 50, 54-59. Retail example applies the notes' capability model.

Capability	Customization rule
Memory	Standardize; no customization.
Integration	Customize for roles.
Insight	Customize for roles AND tasks.
Presentation	Customize for roles, tasks AND individuals.

Source: PDF pp. 88-89.

Feature-to-capability mappings

Benchmarking -> organizational memory. Intelligence -> information integration + insight creation. Convenience -> presentation. Benchmarking examines performance and industry trends; intelligence develops insights from disparate sources; convenience makes output appropriate and user-friendly.

Source: PDF p. 44.

MCQ traps: warehouse/ERP = memory; text/web mining = integration; data mining = insight; OLAP/dashboard = presentation. A lack of current external inputs points to integration. Valid insights that users cannot understand point to presentation.

Cover and recall: name two technologies for each capability, then reproduce the four customization rules without looking.

4. Developing a good BI solution

Know the property, the stage and the development approach.

Quality	Meaning
Alignment	Fits business strategy, processes and context.
Intelligence	Provides new, valuable insight.
Usability	Easy to use.
Accuracy	Based on clean, accurate data.
Connectivity	Interfaces effectively with other IT systems.
Flexibility	Can be modified quickly for changing business conditions.
Portability	Works on different software/hardware platforms.
Scalability	Expands as users and data volume grow.

Source: PDF p. 72.

Development step	Recognize these activities
Plan	Objectives, alignment, scope, deliverables, cost-benefit, infrastructure, staffing and organizational involvement.
Analyse	Model processes, develop metrics, involve users, assess readiness.
Design	Detailed specifications for all four capabilities.
Build	Create the solution or pilot from the design.
Deploy	Test the current iteration, train users/support staff, roll out.
Review	Evaluate satisfaction, what works, what fails and what to improve.

Source: PDF pp. 81-82.

Waterfall	Agile
Strictly sequential; finish a stage before the next.	Iterative; repeat steps, deliver small chunks and build on them.
Comprehensive documentation; contract; planning; process.	Working software; collaboration; flexibility; people.
Described as heavy; steps strictly documented.	Described as light; more flexible.

Source: PDF pp. 77-80.

MCQ traps: more users/data = scalability; another platform = portability; changing business requirements = flexibility; connecting to ERP = connectivity. Choosing metrics = Analyse; specifying capabilities = Design; training and rollout = Deploy; stakeholder satisfaction = Review.

Cover and recall: a solution works on several platforms but cannot exchange data with ERP. Which property is good and which is deficient? Answer: portability is present; connectivity is deficient.

5. Sourcing, users, tools and vendors

Tool origin and solution developer are separate decisions.

Sourcing approach	Who provides tools / development?
Completely in-house	Internal tools; internal development.
Vendor tools, in-house development	Acquired vendor tools, possibly internal tools; internal development.
Joint development	Vendor tools, possibly internal tools; joint development.
Outsourced development with internal tools	Some internal tools; external development.
Complete outsourcing	External tools and external development.

Source: PDF pp. 73-74.

Three vendor/tool choices: strategic vs tactical buying; mega-vendors vs independent vendors; standardization vs multiple options. Strategic buying builds long-term relationships with a few preferred vendors; tactical buying selects the best option for a particular need. Standardization depends on prior BI experience, internal IT competence and resources.

Source: PDF p. 75.

User type	Typical behavior
Powerful users	Analysts or others with a strong understanding of BI tools.
Business users	Information consumers with little BI expertise.
Middle class	Usually consume information; occasionally create reports/queries or perform more analysis.

Too much customization can waste time on tools and produce narrowly useful queries. Too much standardization can restrict users and inhibit BI benefits. Use the capability-specific customization rules on guide page 4.

Source: PDF pp. 86-89.

Vendor category	Ability to execute (ATE)	Completeness of vision (COV)
Leader	Strong	Strong
Challenger	Strong	Limited
Visionary	Limited	Strong
Niche player	Limited	Limited

ATE criteria: product/service, viability, sales execution/pricing, market responsiveness/track record, marketing execution, customer experience, operations. COV criteria: market understanding, marketing/sales/product strategies, business model, industry strategy, innovation, geographic strategy.

Source: PDF pp. 91-93.

Vendor examples in the notes: mega-vendors include Oracle, SAP, IBM and Microsoft; major independent vendors listed include Tableau, SAS, MicroStrategy, Information Builders, TIBCO and Qlik. These are the notes' historical categories, not a current market ranking.

Source: PDF p. 90.

Cover and recall: strong execution but limited vision = Challenger. Limited execution but strong vision = Visionary. Do not swap them.

6. Managing BI: processes and culture

Successful BI needs people, resources and evaluation as well as tools.

Management aspect	Contents
Processes	Business case; organizational support; education; evaluation; development; sourcing.
Infrastructure	Data infrastructure and organizational culture.
Structures	Governance; competency center; degree of centralization.

Source: PDF pp. 100-101, 112, 117.

Business case: justify and sustain resources

- Show organizational benefits to obtain and keep investment.
- Examples: cross-selling/up-selling; customer satisfaction and retention; competitiveness; faster, more accurate regulatory reporting.
- Cross-selling/up-selling: use customer history and characteristics to find additional products to offer.
- Quantitative ROI can be difficult because BI payoffs may be long term and hard to compute; explain the fundamental value proposition too.

Source: PDF pp. 103-105.

Organizational support: reduce resistance

People may doubt the benefits, feel threatened, prefer intuition or resist change. Build support using a solid business case, incremental investment with visible results, roles that show personal benefit, BI champions, and early user involvement to manage expectations.

Source: PDF pp. 106-107.

Education: broader than tool training

Keep current and potential users informed about benefits, functionality and use. Continue learning through communication, collaborative sessions and the BICC. Develop appreciation for data and analysis, not just skill at clicking tool buttons.

Source: PDF pp. 108-109.

Evaluation: compare expectations with outcomes

Use the business-case expectations as a baseline. Include unexpected benefits/costs, solicited and unsolicited user feedback, business-strategy alignment, and effects on customers and suppliers. The evaluation material appears on a slide headed 'Education' in the notes.

Source: PDF p. 110.

Analytics culture

Organizational culture means shared norms and beliefs guiding behavior. BI works better where data and analysis guide decisions. Executives establish the example by seeking facts. Technology + analytics culture supports BI success; culture and BI success reinforce one another.

Source: PDF p. 115.

MCQ trap: buying a tool does not fix resistance to change. Choose the process intervention that fits the problem: resource justification = business case; resistance = support; ongoing understanding = education; benefits against expectations = evaluation.

Cover and recall: explain why an incremental pilot and a BI champion can help an organization that prefers intuition-based decisions.

7. Data quality, governance and centralization

Distinguish the exact defect or responsibility.

Data-quality dimension	Question to ask	Failure example
Existence	Do we have the data?	Required customer data was never collected.
Validity	Is the value in the permitted range/domain?	Rating 9 where only 1-5 is allowed.
Consistency	Do copies in different places agree?	Two systems give different customer postcodes.
Integrity	Are relationships complete across data?	An order points to a missing customer record.
Accuracy	Does the value describe reality correctly?	Recorded delivery date differs from actual delivery.
Relevance	Does this data support the business objective?	Correct data concerns the wrong business question.

Source: PDF pp. 113-114. Failure examples apply the definitions.

Two identical wrong values are consistent but inaccurate. A value can be valid yet inaccurate: an allowed age can still be the wrong age. More records and prettier charts cannot repair poor source data.

BI governance: direction, authority and oversight

Component	Remember
Guiding principles	Beliefs guiding BI vision and goals.
Decision-making bodies	People/groups making decisions, providing input or sponsorship, and their interactions.
Decision areas and rights	Which decisions matter and who may make them: investments, project status, adoption/utilization.
Oversight mechanisms	Formal policies/procedures for implementation and progress evaluation.

Source: PDF pp. 118-119.

BICC = Business Intelligence Competency Center: a relatively permanent cross-functional team supporting effective BI use. It coordinates solutions/tools, shares best practices, supports data stewardship, develops standardized models and provides BI expertise.

Source: PDF pp. 120-121.

Centralization dimension	More centralized	More decentralized
Solution focus	Enterprise-wide	Department-focused
Report development	Reports standard in the solution	Reports mainly user-developed

These are separate dimensions. Department-focused solutions may still provide centrally developed reports. The notes recommend a balanced approach to report development.

Source: PDF p. 122.

Cover and recall: missing cross-dataset links = integrity; contradictory copies = consistency; wrong real-world description = accuracy. Governance allocates authority; the BICC supplies coordinated expertise.

8. BI strategy: problems and roadmap

Align the BI direction with the business direction.

A BI strategy is a roadmap for measuring performance, seeking competitive advantage and understanding customers through data mining and statistics. BI provision should align with corporate strategy.

Source: PDF pp. 126, 131.

Reason strategy is needed	Recognize the problem
Multiple versions of the truth	Different data produces conflicting figures/conclusions; reliable decisions need a shared version.
Unclear business logic	The same term has different definitions, or calculations are named inconsistently.
Personnel overhead	Separate departmental BI efforts can duplicate staffing; a BICC can help.
Programming instead of analysing	Analysts spend time extracting data rather than interpreting results.
Data quality issues	Visualizations and insights use poor data: garbage in, garbage out.

Source: PDF pp. 127-129.

Strategy area	What it decides
1. BI roadmap	Scope/exclusions and alignment with business strategy.
2. Stakeholder management	Key information consumers, critical subject areas and upcoming organizational challenges.
3. Architectural blueprint	BI structure and delivery; development methodology; team funding and reporting lines.
4. Capability improvement	People/skills/resourcing; software and hardware needed for information delivery.
5. Implementation planning	Programme management, release schedules, documentation expectations and BI team involvement.

Source: PDF pp. 130-138.

Infrastructure planning: investigate which software fits the strategic direction, then shape the hardware architecture around it. People may need updated skills or external/new staff.

Source: PDF p. 137.

Additional considerations: how to acquire, validate, store, protect and process required data for accessibility, reliability and timeliness; types of reports and both dimensions of centralization.

Source: PDF p. 140.

MCQ traps: roadmap = scope and alignment; stakeholders = who needs information; blueprint = structure/funding/reporting; improvement = people and infrastructure; implementation = releases and documentation. Conflicting totals concern versions of truth; conflicting meanings concern business logic.

Cover and recall: name the five strategy areas in order and give one decision handled by each. Then explain how poor data can undermine an otherwise impressive visualization.

9. Power Query: the practical workflow

Focus on what each transformation does and the final level of detail.

Power Query / Get & Transform gathers, connects, combines, refines and reshapes data. Sources in the lab include a web table, an Excel workbook and an OData feed. The notes describe OData as an open data-sharing protocol based on REST principles.

Source: PDF pp. 146, 149.

Operation	Effect
Use first row as headers	Promote the selected first row into column names.
Remove Other Columns	Keep selected columns; remove unselected ones.
Replace Values	Substitute a specified value; the lab replaces null with 0.
Filter	Retain/exclude rows using chosen values or conditions.
Applied Steps	Records successive query actions and transformations.
Expand	Bring selected fields from a related/nested table into the main table.
Custom column	Create a calculated value, such as Line Total.
Group By	Aggregate records at the chosen key level.
Merge	Match two queries on selected common key fields.

Source: PDF pp. 147-153.

Sales per product per year: learn this sequence

- Import Products and Orders; expand Order_Details to obtain ProductID, UnitPrice and Quantity.
- Keep the relevant fields. Add Line Total = UnitPrice x Quantity.
- Extract Year from OrderDate and rename that field Year.
- Group by ProductID AND Year. Total Sales = Sum of Line Total.
- Merge Products with Total Sales on the matching ProductID fields.
- Expand Year and Total Sales; sort Total Sales descending to find the largest sales values.
- Load the result as needed; the lab uses connection-only/data-model options, then a worksheet table and PivotTable.

Source: PDF pp. 149-155.

Worked example: product A has two lines in 2025: 10 x 3 = 30 and 8 x 5 = 40. A line in 2026 is 12 x 2 = 24. Grouping by product and year gives A/2025 = 70 and A/2026 = 24. Grouping only by product gives 94 and loses the yearly distinction.

Example applies PDF pp. 150-151.

MCQ traps: Line Total is multiplication, not addition. Sum the line totals, not unit prices. Grouping keys determine the answer's level of detail. Merge creates a related-table column; expand exposes the desired fields. The lab's null-to-zero replacement is a specific exercise step.

Cover and recall: why are ProductID and Year both needed? What is the difference between Merge and Expand?

10. Tableau fields, charts and workspace

Field role and discrete/continuous behavior are separate choices.

Distinction	Meaning
Dimension	Qualitative categories such as names, dates or geography; segments data and affects detail.
Measure	Quantitative numeric values; can be aggregated. Tableau aggregates a measure in a view by default.
Discrete / blue	Separate values; produces headers.
Continuous / green	An unbroken range; produces axes.

A dimension may be continuous, and a measure may be discrete. Blue does not always mean dimension; green does not always mean measure. A continuous field on Columns creates a horizontal axis; on Rows it creates a vertical axis.

Source: PDF pp. 166-167.

Chart	Best recognition cue
Area	Shaded area under a line; accumulated totals over time or stacked lines.
Bar	Compare values across categories; dimension and measure on opposing shelves.
Bullet	Compare a primary performance measure with one or more comparison measures.
Histogram	Shape of a distribution; numeric values grouped into bins/ranges.
Treemap	Nested rectangles; dimensions define structure; measures determine size/color.

Source: PDF pp. 167-177.

Treemap example: Sales on Size and Profit on Color means equal sales can produce equal areas but different colors. For the histogram's per-bar segment percentages, use Percent of Total and edit Compute Using to Cell as the lab specifies.

Source: PDF pp. 175-177.

Workspace feature	Remember
Start page	Connect, Open and Discover panes.
Workbook / sheets	Sheets may be worksheets, dashboards or stories.
Rows / Columns	Place fields to construct the view; Swap exchanges these shelves.
Marks / Color	Change mark type and encode groups or values with color.
Filters	Restrict the data shown; the lab filters Region to focus on West.
Show Me	Suggest chart types suited to the selected fields.
Status bar	Information about the current view, including marks and totals.

Source: PDF pp. 158-165, 168, 171-172.

MCQ traps: histogram = distribution, bar = category comparison, bullet = performance comparison. Discrete/continuous controls headers/axes; dimension/measure controls categorization/measurement. These distinctions are related but not identical.

Cover and recall: what happens when Quantity changes from continuous to discrete? Answer: the axis becomes headers, with green changing to blue.

11. Tableau: combine, connect and clean

Work out rows, keys and grain before choosing a combining method.

Method	Effect / when useful
Union	Append rows vertically; compatible tables have matching columns. Monthly tables -> one longer table.
Join	Match on common fields and combine columns. The number of rows depends on matches and duplicates.
Data blending	Combine a primary source with secondary sources on common fields; useful when joins are unavailable or levels of detail differ, such as daily vs quarterly.

Source: PDF pp. 181-186.

Join type	Rows retained
Inner	Only matching rows from both sides.
Left	All left rows and right matches; missing right values become null.
Right	All right rows and left matches; missing left values become null.
Full outer	Rows from both sides, including unmatched rows; missing-side values become null.

Original examples: left IDs {1,2}, right IDs {2,3}, unique rows: inner -> {2}; left -> {1,2}; right -> {2,3}; full outer -> {1,2,3}. A union of tables with 12, 18 and 20 rows gives 50 rows when nothing is filtered or deduplicated.

Duplicate-key trap: two left rows and three right rows with the same key produce 2 x 3 = 6 matched rows. If left sales are 100 and 200, each repeats three times, so a joined sum becomes 900 rather than the original 300. Check table detail before interpreting totals.

Examples apply PDF pp. 166, 181, 183-186.

Preparation / connection	Remember
Live	Data stays in source; useful for current updates and capable databases; can be slow depending on source.
Extract	Bring data into Tableau's analysis engine; can improve slow-source performance and reduce source query load.
Metadata	Review/rename/hide columns, change types or geographic roles; these changes do not modify underlying data.
Data-source filter	Reduce the data included in the data source.
Split / custom split	Separate a string using a delimiter or repeated pattern, e.g. location and employee ID.
Data Interpreter	Detect titles, notes, footers, empty cells and embedded tables. Check the result and do further cleaning if needed.
Pivot	Turn original column names and values into Pivot field names / Pivot field values; the lab renames these Date / Resolved Incidents.

Source: PDF pp. 186-193.

Final check: explain why a join can increase row count; why a union differs from a join; why live differs from extract; and how pivot differs from splitting a field. Then return to the checklist on guide page 1.